Graphic Era HILL UNIVERSITY

Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

End Term (Odd) Semester Examination December 2024

Roll no.....

(ME)

Name of the Course and semester: B.Tech V semester
Name of the Paper: Operation Research
Paper code: TME 505
Time: 3 hour

Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks)

a. Write down the characteristics and advantages of operation research. (CO1)

b. Use simplex method to solve the following LP problem (CO1)

Maximize $Z = x1+x2+3x3$
Subject to $3x1+2x2+x3 \leq 3$
$2x1+x2+2x3 \leq 2$
$x1, x2 \geq 0$

c. Minimize $Z=3x1+2x2$ solve by graphically. (CO1)

Subject to $5x1+x2 \geq 10$
$x1+x2 \geq 6$
$x1+4x2 \geq 12$
$x1,x2, \geq 0$

Q2. (2X10=20 Marks)

a. Write down in brief about the different methods used in Assignment and Transportation. (CO2)

b. Solve by Using VAM Method: (CO2)

Origin/Destination	D1	D2	D3	D4	Supply
O1	11	13	17	14	250
O2	16	18	14	10	300
O3	21	24	13	10	400
Demand	200	225	275	250	.

c. The Processing time in hours for the jobs when allocated to different machines is indicated below. Assign the machines for the jobs so that the total processing time is minimum: (CO2)

	M1	M2	M3	M4	M5
J1	9	22	58	11	19
J2	43	78	72	50	63
J3	41	28	91	37	45
J4	74	42	27	49	39
J5	36	11	57	22	25

Q3. (2X10=20 Marks)

a. Write down the difference between PERT and CPM. (CO3)

b. A project schedule has the following characteristics. Draw the network diagram and find the critical path. (CO3)

Graphic Era HILL UNIVERSITY

Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

End Term (Odd) Semester Examination December 2024

Activity	1-2	1-3	2-3	2-4	3-4	4-5
Time(days)	20	25	10	12	6	10

c. A project has the following activities and other characteristics: Find out expected project length (CO3)

Activity	1-2	1-3	1-4	2-5	3-5	4-6	5-6
a	1	1	2	1	2	2	3
m	1	4	2	1	5	5	6
b	7	7	8	1	14	8	15

Q4. (2X10=20 Marks)

a. In a public telephone booth the arrivals are on the average 15 per hour. A call on the average takes 3 minutes. If there is just one phone, find (i) the expected number of callers in the booth at any time (ii) the proportion of the time the booth is expected to be idle? (CO4)

b. Write short notes on game theory and queuing theory. (CO4)

c. Reduce the following game by dominance and find the game value. (CO4)

		Player B			
		I	II	III	IV
Player A	I	3	2	4	0
	II	3	4	2	4
	III	4	2	4	0
	IV	0	4	0	8

Q5. (2X10=20 Marks)

a. What do you understand by sequencing? Write down the assumptions made in it. (CO5)

b. Find out sequence using SPT rule and find out average tardiness of the given problem: (CO5)

Job	A	B	C	D	E	F	G
P.T.	4	12	2	11	10	3	6
D.D.	20	30	15	16	18	5	9

c. Find out total elapsed time and idle time of each machine: (CO5)

Job	1	2	3	4	5	6	7
Machine A	3	12	15	6	10	11	9
Machine B	8	10	10	6	12	1	3

Note For the question paper setters:

- Question paper should cover all the COs of the course.
- Please specify COs against each question.